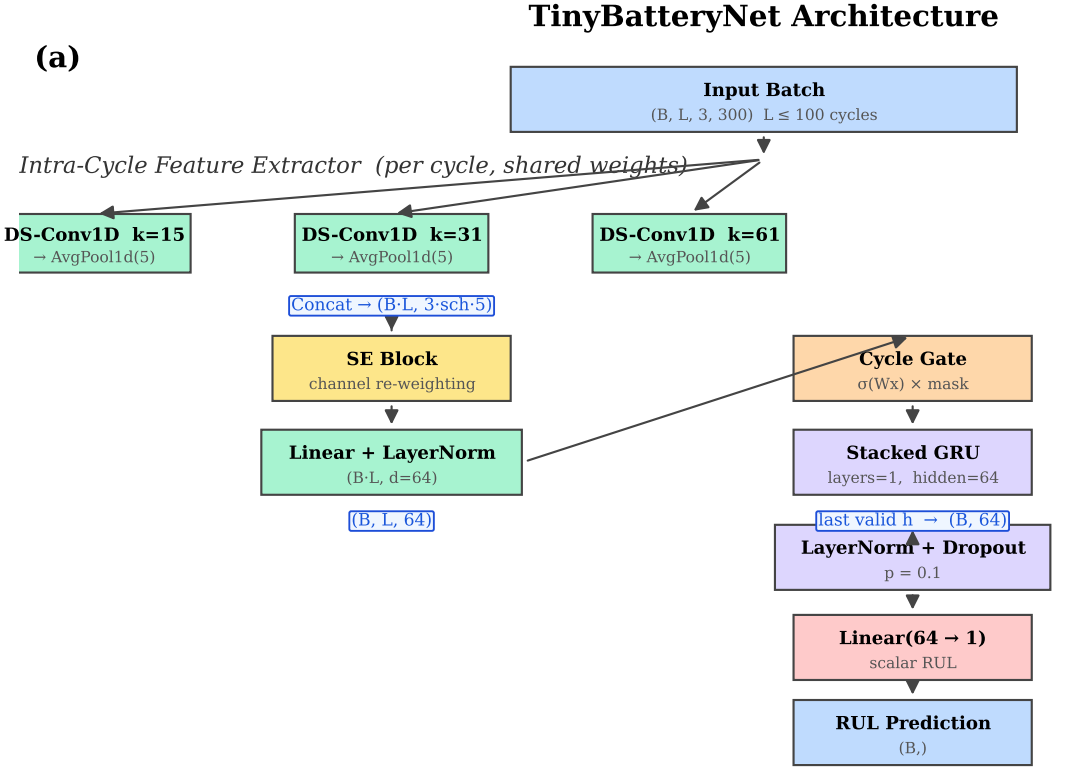
TinyBatteryNet Architecture
(a)
Input Batch
(B, L, 3, 300) L ≤ 100 cycles
Intra-Cycle Feature Extractor (per cycle, shared weights)
DS-Conv1D k=15
→ AvgPool1d(5)
DS-Conv1D k=31
→ AvgPool1d(5)
DS-Conv1D k=61
→ AvgPool1d(5)
Concat → (B·L, 3·sch·5)
SE Block
channel re-weighting
Linear + LayerNorm
(B·L, d=64)
(B, L, 64)
Cycle Gate
σ(Wx) × mask
Stacked GRU
layers=1, hidden=64
last valid h → (B, 64)
LayerNorm + Dropout
p = 0.1
Linear(64 → 1)
scalar RUL
RUL Prediction
(B,)